

Problem A. I Hate Minimums

Input file: standard input
 Output file: standard output
 Balloon Color: Pink

You are given an array a of n integers and q queries. Each query is defined by two integers l and r , representing a segment of the array.

For each query, your task is to count the number of **subarrays** $[a_i, a_{i+1}, \dots, a_j]$ such that:

- $l \leq i \leq j \leq r$ (i.e., the subarray is fully contained within the interval $[l, r]$), and
- The **minimum value** of the selected subarray is **equal** to the minimum value of the elements **outside** the subarray but **within** the query range $[l, r]$.

In other words, you are to compare the minimum of a subarray inside $[l, r]$ with the minimum of the *remaining elements* in $[l, r]$ that are not in that subarray.

Note: If the subarray covers the entire range $[l, r]$, then the remaining elements are empty, and their minimum is considered to be 0.

Input

The input consists of:

- A single integer n ($1 \leq n \leq 10^5$) — the size of the array a .
- An array a of n positive integers, where each element satisfies $1 \leq a[i] \leq 10^9$.
- A single integer q ($1 \leq q \leq 10^5$) — the number of queries.
- q queries, each in the form of two integers l, r ($1 \leq l \leq r \leq n$).

Output

output answer for each query

Example

standard input	standard output
9	3
1 3 2 2 2 2 3 2	5
5	0
6 8	0
5 7	5
2 3	
9 9	
4 6	

Handwritten annotations for the example queries:

- Query 1: $l=1, r=9$. Subarray $[2, 3, 2]$ is highlighted. The remaining elements are $[1, 2, 2, 2, 2, 3, 2]$. The minimum of the subarray is 2, and the minimum of the remaining elements is 1. Since $2 \neq 1$, this subarray is not counted. The correct answer is 3, corresponding to subarrays $[2, 2]$, $[2, 3]$, and $[3, 2]$.
- Query 2: $l=6, r=8$. Subarray $[7]$ is highlighted. The remaining elements are $[6, 8]$. The minimum of the subarray is 7, and the minimum of the remaining elements is 6. Since $7 \neq 6$, this subarray is not counted. The correct answer is 0.
- Query 3: $l=5, r=7$. Subarray $[2, 3, 2]$ is highlighted. The remaining elements are $[5, 7]$. The minimum of the subarray is 2, and the minimum of the remaining elements is 5. Since $2 \neq 5$, this subarray is not counted. The correct answer is 5, corresponding to subarrays $[5]$, $[7]$, $[5, 7]$, $[2, 2]$, and $[2, 3]$.
- Query 4: $l=2, r=3$. Subarray $[2, 3]$ is highlighted. The remaining elements are $[2]$. The minimum of the subarray is 2, and the minimum of the remaining elements is 2. Since $2 = 2$, this subarray is counted. The correct answer is 1.
- Query 5: $l=9, r=9$. Subarray $[9]$ is highlighted. The remaining elements are empty. The minimum of the subarray is 9, and the minimum of the remaining elements is 0. Since $9 \neq 0$, this subarray is not counted. The correct answer is 0.
- Query 6: $l=4, r=6$. Subarray $[2, 3, 2]$ is highlighted. The remaining elements are $[4, 6]$. The minimum of the subarray is 2, and the minimum of the remaining elements is 4. Since $2 \neq 4$, this subarray is not counted. The correct answer is 0.

Handwritten notes on the right side of the page:

1 2 3
 S S S
 E E E
 E E E

Problem B. Yassin And Strings

Input file: standard input
 Output file: standard output
 Balloon Color: Orange

You are given n strings which represent a magical dictionary where the i_{th} string is d_i and a target string s .

Help Yassin to count the number of ways we can concatenate some (possibly one) strings from the dictionary in any order and any number of times such that they form the given string s .

For example if the strings in the dictionary are $\{ a, a, bc \}$ then you can form the following strings :

- $d_1 + d_2 + d_3 + d_1 = aabca$
- $d_2 + d_1 + d_3 = aabc$
- $d_1 + d_3 + d_2 = abca$

Since the answer might be large so print the answer module $10^9 + 7$.

Input

The first line contains an integer (n) ($1 \leq n \leq 10^5$) — the number of words in the magical dictionary.

Each of the next n lines contains a non-empty string d_i consisting of lowercase English letters.

The next line contains a single integer m the size of the string s .

The final line contains a non-empty string s of length at most 10^5 , the long word Yassin wants to make it.

It is guaranteed that the summation of sizes of the strings d_i not exceeding 10^5

Output

Print one integer, the answer of the problem module $10^9 + 7$.

Example

standard input	standard output
5 i is show speed how 10 ishowspeed	2



Problem C. Unstable Draw Game

Input file: standard input
Output file: standard output
Balloon Color: Purple

Anne and Brian are playing a game with a deck of N cards. The rules are as follows:

- Anne always plays first.
- On each turn, a player must remove 1, 2, or 3 cards from the deck.
- A player must not remove the **same number of cards** as the previous player.
- If the **absolute difference** between the number of cards removed by the current player and the previous player is 2 (a prime number), the current player **immediately loses**.
- The player who removes the last card wins, as long as no rules are violated.
- **On the first move only**, Anne can choose any valid move (1, 2, or 3) without restrictions. That is, Rules 3 and 4 do not apply to the first move.

Both players play optimally. Determine who will win the game.

Input

The first line contains a single integer T ($1 \leq T \leq 10^5$) — the number of test cases.

Each test case contains one integer N ($1 \leq N \leq 10^8$) — the initial number of cards in the deck.

Output

For each test case, print a single line containing the name of the winner: Anne or Brian.

Assume that both players play optimally.

Example

standard input	standard output
1 7	Anne

Note

Both players play optimally.

A player immediately loses if their move violates Rule 4.

The game always ends with one winner.



Problem D. Path of the Katakets

Input file: standard input
Output file: standard output
Balloon Color: Light Blue

You are given a tree with n nodes, where each node is assigned a lowercase English letter.

You are also given a list of m unique strings.

You have to process q queries. In each query, you are given two nodes u and v . Determine whether the string formed by concatenating the characters on the simple path from u to v exists in the given list of strings.

Note that since the input graph is a tree, there is a unique simple path between any two nodes.

It is **guaranteed** that the sum of n across all test cases does not exceed 6×10^5 , the sum of q across all test cases does not exceed 6×10^5 , and the total length of all m strings across all test cases does not exceed 4×10^5 .

Input

The first line contains a single integer t ($1 \leq t \leq 1000$) — the number of test cases.

Then for each test case:

- The first line contains two integers n and m ($1 \leq n \leq 5 \times 10^5$, $1 \leq m \leq 10^4$) — the number of nodes and the number of strings.
- Each of the next $n - 1$ lines contains two integers u and v ($1 \leq u, v \leq n$) — an edge of the tree.
- The next line contains a string of length n — the i -th character is the letter on node i .
- Each of the next m lines contains one string — the list of strings you are to search for.
- The next line contains an integer q ($1 \leq q \leq 3 \times 10^5$) — the number of queries.
- Each of the next q lines contains two integers u and v ($1 \leq u, v \leq n$) — the endpoints of a query path.

Output

For each query, print “YES” (without quotes) if the path string is present in the given list of strings, otherwise print “NO”.

Example

standard input	standard output
1	YES
6 3	
1 2	
1 3	
3 6	
2 4	
2 5	
acbdda	
acd	
bad	
abacd	
1	
6 5	

Problem E. Distinct Palindromes

Input file: standard input
 Output file: standard output
 Balloon Color: Light Pink

You are given T test cases; in each test case, you have an initially empty string s .
 You need to process q queries. Each query is one of the following three types:

- 1 x — Let $f(b)$ be the number of integers c with $1 \leq c \leq b$ and $\gcd(b, c) = 1$. Compute $y = f(|s| + x) \bmod 26$, where $|s|$ is the length of the current string. Take the $(y + 1)$ -th lowercase letter ($a = 1, b = 2, \dots, z = 26$) and append it to s .
- 2 — If the string is non-empty, remove its last character. Otherwise, do nothing.
- 3 — Output the number of **distinct non-empty palindromic substrings** in the current string s .

Input

The first line contains an integer T — the number of test cases ($1 \leq T \leq 10^5$).

Then, for each test case:

The first line contains an integer q — the number of queries in this test case ($1 \leq q \leq 2 \times 10^6$).

Each of the next q lines describes a query, in one of the following formats:

1 x — a query of type 1, where $1 \leq x \leq 10^7$

2 — a query of type 2

3 — a query of type 3

The total number of queries over all test cases does not exceed 2×10^6 .

Output

For each query of type 3, output a single line containing the number of distinct palindromic substrings in the current string s .

Example

standard input	standard output
1	6
9	5
1 1	
1 2	
1 3	
1 4	
1 5	
1 6	
3	
2	
3	

Note

There is at least one query of type 3 in the input.

Problem F. Largest subsequence

Input file: standard input
 Output file: standard output
 Balloon Color: Rose Red

Given two strings x and y , find the largest lexicographical subsequence of x that is also an anagram of y . A

If no such subsequence exists, return -1. fix

An anagram of y is any string that uses the same characters with the same frequency, but possibly in a different order. O(n log n)

Input

The input begins with a single integer t ($1 \leq t \leq 10^5$), indicating the number of test cases.

For each test case, the input consists of:

A line with two integers n and m ($1 \leq n, m \leq 10^5$), representing the lengths of strings x and y , respectively.

A line containing the string x of length n .

A line containing the string y of length m .

It is guaranteed that the sum of n and the sum of m over all test cases does not exceed 10^5 .

Output

Print the lexicographically largest subsequence of s that is an anagram of t . If no such subsequence exists, print -1

Example

standard input	standard output
3	cba
9 3	ab
abcabcaba	-1
abc	
5 2	
aacab	
ba	
4 2	
aaba	
bb	

Problem G. Maximize AND Count

Input file: standard input
 Output file: standard output
 Balloon Color: Green

You are given an initially empty multiset S . You need to process Q queries of three types:

1. **Add** element x to S .
2. **Remove** element x from S .
3. **Find** $\max_{a \in S}(x \& a)$ and the number of elements in S that give this maximum value.

Input

The first line contains an integer Q ($1 \leq Q \leq 5 \times 10^5$) — the number of queries. Each of the next Q lines contains two integers t and x ($1 \leq t \leq 3$, $1 \leq x \leq 10^5$), where t is the query type:

- If $t = 1$, add the element x to S .
- If $t = 2$, remove element x from S . It is guaranteed that x exists in the multiset when this operation is performed.
- If $t = 3$, find $\max_{a \in S}(x \& a)$ and the number of elements a_i in S that achieve this maximum value.

Note: If the multiset S is empty and a query of type 3 is performed, the output should be 0 0.

Output

For each query of type 3, output two space-separated integers: $\max_{a \in S}(x \& a)$ and the number of elements in the set S that give this maximum value when performing bitwise AND with x .

Examples

standard input	standard output
7	2 1
1 1	1 2
1 2	5 1
1 5	
3 3	
2 2	
3 3	
3 5	
10	10 1
1 10	8 1
1 2	2 1
1 8	0 2
3 10	
2 10	
3 10	
1 4	
3 3	
2 4	
3 5	

Problem H. Welcome to school

Input file: standard input
Output file: standard output
Balloon Color: Dark Green

Every morning, a teacher and a student come to class. You are given the hours when the student arrives and the teacher arrives (both expressed as integers on a 12-hour clock).

If the student arrives strictly before the teacher, the student says "YES"; otherwise, the student says "NO".

Input

The input consists of two lines.

The first line contains a single integer a ($1 \leq a \leq 12$) — the hour when the student arrives.

The second line contains a single integer b ($1 \leq b \leq 12$) — the hour when the teacher arrives.

Output

Output

Print "YES" if the student arrives strictly before the teacher, and "NO" otherwise.

Examples

standard input	standard output
3 9	YES
5 5	NO

Problem I. Abstract

Input file: standard input
 Output file: standard output
 Balloon Color: Navy Blue

Abouraya is studying abstract algebra and came up with three new operators on functions, denoted by Δ , ∇ , and $\circ$.

- Δ and ∇ take two functions and return a new function.
- $\circ$ takes one function and returns a new function.

These operators follow the rules:

1. $f\nabla g = g\nabla f$
2. $f\Delta g = g\Delta f$
3. $f\Delta(g\nabla h) = (f\Delta g)\nabla(f\Delta h)$
4. $\circ(f\nabla g) = (\circ f)\nabla(\circ g)$
5. $\circ(f\Delta g) = ((\circ f)\Delta g)\nabla(f\Delta(\circ g))$

Abouraya starts with m functions: $[f_1, f_2, \dots, f_m]$

First, he combines them with the Δ operator: $[f_1\Delta f_2\Delta \dots \Delta f_m]$

Then, he applies the $\circ$ operator exactly n times to the whole expression.

It can be shown that after expansion, the final expression becomes a ∇ -combination of several *groups*:

- Inside each group, the functions are connected by Δ .
- In every group, each f_i appears with some number of $\circ$ applied to it.

For example, a group may look like:

$$[(\circ(\circ f_1))\Delta(\circ f_2)\Delta f_3]$$

Here, f_1 has 2 applications of $\circ$, f_2 has 1, and f_3 has 0.

Abouraya now gives you an array $a = [a_1, a_2, \dots, a_m]$, where a_i is the number of times $\circ$ is applied to f_i in some group.

Your task is to determine how many times this exact group appears in the final expression. Since the number can be large, output it modulo $10^9 + 7$.

Input

The first line contains a single integer t ($1 \leq t \leq 10^4$) — the number of test cases.

Each test case consists of two lines:

- The first line contains two integers n, m ($1 \leq n, m \leq 10^5$).
- The second line contains m integers $a_1, a_2, \dots, a_m$ ($0 \leq a_i \leq 10^5$).

It is guaranteed that the sum of all m across test cases does not exceed 10^5 .

Output

For each test case, print a single integer — the number of times the given group appears, modulo $10^9 + 7$.

Example

standard input	standard output
3	1
2 1	2
2	0
2 2	
1 1	
3 2	
1 1	

Note

Example 1:

If $m = 1$, $n = 2$, and $a = [2]$, the final expression has exactly one group:

$[\circ(\circ f_1)]$

So the answer is 1.

Example 2:

If $m = 2$, $n = 2$, and $a = [1, 1]$, the group $(\circ f_1)\Delta(\circ f_2)$ appears 2 times in the final expression.

Problem J. Gad and primes

Input file: standard input
 Output file: standard output
 Balloon Color: Blue

You are given an array A of N integers where $1 \leq N \leq 10^5$ and each element satisfies $1 \leq a_i \leq 3$.

Your task is to rearrange (permute) the elements of A so that the sum of **every non-empty contiguous subarray** is a prime number.

If no such rearrangement exists, output -1 . Otherwise, output any valid rearrangement.

A **subarray** of an array is a contiguous segment $A[l], A[l+1], \dots, A[r]$ for some $1 \leq l \leq r \leq N$.

A **prime number** is an integer $p \geq 2$ that has exactly two positive divisors.

Input

The first line contains a single integer t — the number of test cases. For each test case:

- The first line contains an integer N ($1 \leq N \leq 10^5$).
- The second line contains N integers $a_1, a_2, \dots, a_N$ ($1 \leq a_i \leq 3$).

It is guaranteed that the sum of N across all test cases does not exceed 10^5 .

Output

For each test case, print a single line:

- Print -1 if no valid rearrangement exists.
- Otherwise, print N integers forming a rearrangement of the given multiset such that the sum of every non-empty contiguous subarray is prime.

Example

standard input	standard output
3	-1
2	-1
2 2	2 3
5	
2 1 2 3 2	
2	
2 3	

Note

Any valid rearrangement is acceptable. If multiple solutions exist, you may print any of them.

Problem K. Selling The Contribution

Input file: standard input
Output file: standard output
Balloon Color: Black

Dots has realized that money is more important than extra slots in the ECPC, so he is planning to sell the contribution.

The contribution consists of n problems, each with specific pricing constraints. The site Dots plans to sell to has a strict reputation system, and the following rules apply to each problem:

- If the i -th problem is sold for a price greater than a_i but less than or equal to b_i , Dots gains **negative reputation**.
- If the i -th problem is sold for a price greater than b_i , the problem is **declined**; this does not affect Dots' reputation, but he does not gain any money.
- Selling any problem for a price less than or equal to a_i is always safe and does not affect Dots's reputation.

If Dots accumulates more than k instances of negative reputation, he will be banned from the site.

Your task is to determine the price of each problem such that Dots maximizes his total gains while avoiding a ban.

If there are multiple solutions, print any of them.

Input

The first line contains t ($1 \leq t \leq 1000$) — the number of test cases.

- The first line of each test contains two integers n and k ($1 \leq n \leq 10^5, 0 \leq k \leq n$) — the number of problems and the maximum number of negative reputation instances Dots can afford without being banned.
- The second line of each test contains n integers $a_1, a_2, \dots, a_n$ ($1 \leq a_i < 10^9$) — the lower bound for each problem's price.
- The third line of each test contains n integers $b_1, b_2, \dots, b_n$ ($a_i < b_i \leq 10^9$) — the problematic range for each problem's price.

The sum of n over all test cases does not exceed 10^5

Output

For each test, print n integers — the price for each problem such that Dots maximizes his gains without getting banned.

Example

standard input	standard output
1	2 9 8 6 2
5 3	
2 3 4 2 2	
5 9 8 6 3	

Problem L. The Cursed Tomb of Pharaoh Amun-Ra

Input file: standard input
Output file: standard output
Balloon Color: Sand White

Deep in the Egyptian desert, the legendary treasure of Pharaoh Amun-Ra was stolen by the cunning thief Khepri the Scarab. The pharaoh's royal guards chased him to an ancient tomb, where they found a massive stone door covered in mysterious inscriptions. To open it, they had to solve the tomb's final riddle. They found a word puzzle on the door:

"Answer wisely, or be trapped forever! If the words share an ending, say YES, but know it's not enough, I also want you to tell me the LCS (longest common suffix). If no common suffix binds them, say NO, Fail... and the scarab's curse will strike!"

Unfortunately, the guards answered the first puzzle wrong and would be trapped forever unless you helped them. Can you solve this puzzle and save the Amun-Ra treasure?

Input

The first line contains n ($2 \leq n \leq 10^4$) - the number of words on the door.

The Second line contains $s_1, s_2, \dots, s_n$ words with size m ($2 \leq m \leq 100$).

Words aren't necessarily the same size, and all their characters are lower.

Output

See if there is a longest common suffix between the given words or not:

If there is LCS, print "YES" without quotes in a single line, followed by LCS.

If not, print "NO".

Note that LCS can be a single character or word.

Examples

standard input	standard output
2 is vs	YES s
4 information civilization initialization negotiation	YES ation
3 three add are and	NO

Problem M. Sorting Crates in the Warehouse

Input file: standard input
Output file: standard output
Balloon Color: Red

In a warehouse, n crates are placed in a row. Each crate has a weight w_i . The crates need to be rearranged in non-decreasing order by weight.

A robotic arm can only swap adjacent crates, and every time a swap is performed, the “stability flag” of both crates toggles between Stable and Unstable.

Initially, all crates are Stable. Is it possible to sort the crates such that all are Stable again at the end?

Input

The first line contains a single integer t ($1 \leq t \leq 50$) — the number of test cases.

Each test case consists of two lines:

The first line contains an integer n ($1 \leq n \leq 10^5$) — the number of crates.

The second line contains n integers $w_1, w_2, \dots, w_n$ ($1 \leq w_i \leq 10^9$) — the weights of the crates.

The sum of n over all test cases does not exceed 10^5 .

Output

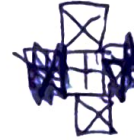
For each test case, print “YES” if it’s possible to sort the crates and have all flags stable. Otherwise, print “NO”.

Examples

standard input	standard output
3 4 4 3 2 5 4 3 3 2 2 5 1 2 3 5 4	YES YES NO
2 3 3 1 2 4 4 1 3 2	NO NO

Problem N. Geo!Geo!Geo!!!

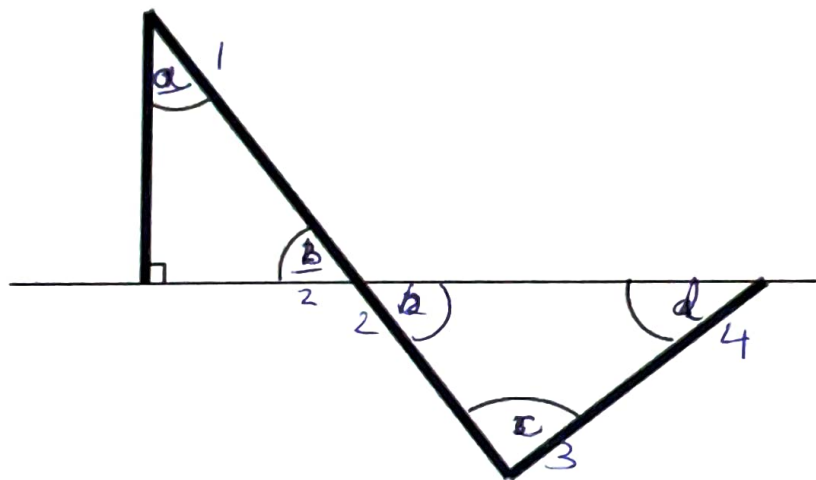
Input file: standard input
 Output file: standard output
 Balloon Color: Yellow



In a geometric shape with four angles 1, 2, 3, and 4, you are given the values of two of them:

One of the angles among 1 or 2, and one of the angles among 3 or 4.

Determine the values of all the remaining angles, or state that it is impossible to form a valid shape with the given angles.



Input

The first line contains a single integer t ($1 \leq t \leq 10^4$) — the number of test cases. *for loop*

Each test case consists of two lines:

The first line contains two integers i and a ($i \in \{1, 2\}$, $0 \leq a \leq 90$) — the index and value of the first known angle.

The second line contains two integers j and b ($j \in \{3, 4\}$, $0 \leq b \leq 180$) — the index and value of the second known angle.

Output

For each test case, print four integers — the values of angles 1, 2, 3, and 4 in this exact order.

If it's impossible to determine valid angles for the shape, print -1.

Example

standard input	standard output
2	60 30 60 90
2 30	-1
4 90	
1 90	
3 60	

STILL
 $[a, b, c]$

Problem O. Unusual Sort

Input file: standard input
Output file: standard output
Balloon Color: Brown

Yousef has given you an array a of size n and a positive integer x . He wants to know if the array can be sorted in any number of operations.

In one operation, you can:

- Swap a_i with a_j if the difference between i and j is equal to x (i.e, $|i - j| = x$).

Help Yousef determine whether it's possible to make the array sorted in any number of operations.

Input

Each test contains multiple test cases. The first line contains a single integer t ($1 \leq t \leq 10^5$) — the number of test cases. The description of the test cases follows.

The first line of each test case contains two integers n and x ($1 \leq n, x \leq 2 \times 10^5$) — the size of the array and the allowed difference, respectively.

The second line of each test case contains n integers $a_1, a_2, \dots, a_n$ ($0 \leq a_i \leq 2 \times 10^5$) — the elements of the array.

It is guaranteed that the sum of n over all test cases does not exceed 2×10^5 .

Output

For each set of input data, output "YES" if a can be sorted, and "NO" otherwise. You may output in any case.

Example

standard input	standard output
3	NO
3 3	YES
5 4 1	YES
6 2	
3 2 0 4 5 10	
1 2	
5	

Note

In the first test case, no pairs of indices have difference equal to x , so we can't do any operations and the answer is "NO".

In the second test case, we can swap a_1 with a_3 . The array will become $[0, 2, 3, 4, 5, 10]$, so the answer is "YES".

In the third test case, there is only one element, so it is already sorted.